Dataset Overview

The dataset used in this project is the HMS Harmful Brain Activity Classification Dataset, publicly available through the Kaggle competition platform. The dataset can be accessed at:

https://kaggle.com/competitions/hms-harmful-brain-activity-classification

The dataset was developed by Dr. Westover and researchers from the Massachusetts General Hospital and Harvard Medical School Department of Neurology. It was released to support the development of automated algorithms for detecting harmful brain activity patterns from electroencephalogram (EEG) recordings. The development cohort used for model training and validation contains EEG recordings from 1,950 patients (mean age 47.73 years, SD 25.52), including 963 female patients. Expert neurologists annotated approximately 106,800 EEG segments belonging to six target classes: Seizure (SZ), Lateralized Periodic Discharges (LPD), Generalized Periodic Discharges (GPD), Lateralized Rhythmic Delta Activity (LRDA), Generalized Rhythmic Delta Activity (GRDA), and Other. The dataset contains 20,933 seizure segments, 14,856 LPD segments, 16,702 GPD segments, 16,640 LRDA segments, 18,861 GRDA segments, and 18,808 Other segments.

The development data is divided into two subsets:

Dataset 1: EEG recordings from 1,794 patients, containing 11,900 EEG recordings and 68,854 EEG segments annotated by 1–7 experts.

Dataset 2 (High-Quality Annotation Subset): EEG recordings from 1,063 patients, containing 5,939 EEG recordings and 39,946 EEG segments annotated by 10–28 experts. This dataset provides a large-scale, clinically validated benchmark for developing automated harmful brain activity detection systems.